DOMINO TILINGS OF ODD-HEIGHT RECTANGLES UP TO SYMMETRY

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Abstract. We count unlabelled domino tilings of odd-height rectangles of even width up to rectangle symmetries. For three rows, a block decomposition and a half-rectangle construction give the generating function recorded in OEIS A232622, its seventh-order recurrence, and all initial values. We also give stabilizer and asymptotic formulas, an odd-height transfer expression, and an explicit five-row polynomial certificate. The three-row main counting theorem has Lean M1–M6 evidence based on actual tiling-orbit cardinalities. This edition explains the state induction and the four coefficients of the compressed recurrence in detail. The formal coverage reported here is limited to M1–M6; external semantic review and bibliographic priority remain unresolved.

Scope of this edition. Theorem 1.1 and its three-row half-rectangle dependencies have the M1–M6 formal evidence described in Section 7. This edition reports that fixed scope, rather than a certification of all results in the manuscript. Independent mathematical and semantic review remain outstanding, and no bibliographic priority is claimed.

1. Introduction

Let $\mathcal{T}_n$ be the set of domino tilings of a $3 \times 2n$ rectangle, where $n \geq 1$. A domino covers two unit cells sharing an edge. Both orientations are allowed, the dominoes are unlabelled, and no tatami restriction is imposed. Thus a tiling is a set of unordered pairs of adjacent cells. Number rows from top to bottom and columns from left to right. The maps

$$h(r, c) = (4 - r, c), \qquad v(r, c) = (r, 2n + 1 - c), \qquad \rho = hv$$

are reflection in the horizontal axis, reflection in the vertical axis, and rotation through 180°, respectively. Since $3 \neq 2n$, the full symmetry group of the rectangle is $G = \{e, h, v, \rho\} \cong C_2 \times C_2$. Write $a(n) = |\mathcal{T}_n/G|$. We set $\mathcal{T}_0$ to consist of a single empty tiling, fixed by all four abstract group elements, so that $a(0) = 1$.

Mathar [2, Table 10, Conjecture 6] proposed a generating function for these symmetry classes. The equivalent formula in OEIS A232622 [4] is accompanied by a conjectured recurrence of order seven. We give two derivations, with the M1–M6 formal evidence following the half-rectangle route. The first uses a complete classification of vertically indecomposable blocks and the action of G on words in these blocks. The second counts symmetric tilings directly from partial tilings of a half-rectangle. The latter construction also yields a uniform formula for any fixed odd height.

Transfer matrices and indecomposable tilings are established tools in fixed-width enumeration. In particular, Mathar [3, Sections 2.1–2.3] gives the ordinary three-row generating function and the corresponding indecomposable-block generating function. We rederive the required block classification and determine the symmetry action explicitly. DeFord [1, Theorem 6.1] establishes a general recurrence framework for rectangular tilings counted up to symmetry and illustrates it for two-row domino tilings in Section 6.2.1. Thus rationality at fixed height is established background; our odd-height formula specifies the boundary conditions and gives an explicit five-row evaluation. None of our combinatorial arguments

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assumes the target recurrence or a finite list of coefficients. The source entry was checked on October 25, 2026; its conjecture label does not establish that no equivalent proof appears elsewhere. No claim of bibliographic priority is made here.

Reading the widths and counts. The exponent n always records half the full width $2n$. In Section 3, $q_w(S)$ counts a partial rectangle of actual width w, while the induction index j groups widths $2j$ and $2j+1$. Thus $\alpha_j = q_{2j}(\varnothing) = d_j$, whereas $\beta_j = q_{2j}(\{1,2\})$ counts one terminal state. The full transfer uses row vectors; its two-coordinate compression in (3.5) uses column vectors.

1.1. **Main result.** Let

$$d_n = |\mathcal{T}_n|, \qquad h_n = |\operatorname{Fix}(h)|, \qquad v_n = |\operatorname{Fix}(v)|, \qquad r_n = |\operatorname{Fix}(\rho)|.$$

Their ordinary generating functions, with x marking the half-width n, are denoted by D, H, V, R. All generating-function identities are interpreted in $\mathbb{Q}[[x]]$.

Theorem 1.1. *The number of domino tilings of a $3 \times 2n$ rectangle up to rectangle symmetries has generating function*

$$A(x) := \sum_{n\geq 0} a(n)x^n = \frac{1 - 3x - 4x^2 + 10x^3 - 4x^5 + x^6}{(1-x)(1-4x+x^2)(1-4x^2+x^4)}. \tag{1.1}$$

Consequently,

$$\begin{aligned} a(n) = {} & 5a(n-1) - a(n-2) - 19a(n-3) + 19a(n-4) \\ & + a(n-5) - 5a(n-6) + a(n-7), \qquad n \geq 7, \end{aligned} \tag{1.2}$$

with initial values

$$(a(0), \ldots, a(6)) = (1, 2, 5, 14, 46, 156, 561).$$

Multiplying the numerator and denominator in (1.1) by -1 gives the form displayed in A232622. Sections 2 and 3 give the two proofs. Section 4 determines the exact stabilizers and the asymptotic proportion of symmetric tilings. Sections 5 and 6 treat arbitrary odd height and the explicit five-row case.

Lemma 1.2 (Burnside's formula)**.** *For every $n \geq 0$,*

$$a(n) = \frac{d_n + h_n + v_n + r_n}{4}. \tag{1.3}$$

Proof. Count the pairs $(g, T) \in G \times \mathcal{T}_n$ with $gT = T$:

$$\sum_{g\in G} |\operatorname{Fix}(g)| = \sum_{T\in\mathcal{T}_n} |\operatorname{Stab}(T)|.$$

By the orbit–stabilizer formula, each orbit O contributes $|O| \cdot 4/|O| = 4$ to the right-hand side. Division by four proves the assertion. This is the usual orbit-counting lemma; see [5, Chapter 7, Lemma 7.2]. The argument does not require a faithful action and includes $n = 0$. □

2. Indecomposable blocks

2.1. **Column states.** Consider a three-row rectangle of arbitrary width w. At the cut between columns j and $j+1$, let $S_j \subseteq \{1,2,3\}$ be the set of rows crossed by a horizontal domino. Set $S_0 = S_w = \varnothing$. If a column has incoming state p and outgoing state q, then the transition is legal precisely when

$$p \cap q = \varnothing, \qquad \{1,2,3\} \setminus (p \cup q) \in \{\varnothing, \{1,2\}, \{2,3\}\}. \tag{2.1}$$

Indeed, the cells in p are covered from the left, those in q are covered towards the right, and the remaining cells must be paired vertically. Their vertical completion is unique. Table 1 lists every transition.

Lemma 2.1. *Tilings of a $3 \times w$ rectangle are in bijection with legal state paths of length w from $\varnothing$ to $\varnothing$.*

TABLE 1. The complete three-row transition table.

Input	Outputs	Input	Outputs
$\varnothing$	$\{1\},\{3\},\{1,2,3\}$	$\{1,2\}$	$\{3\}$
$\{1\}$	$\varnothing,\{2,3\}$	$\{1,3\}$	$\{2\}$
$\{2\}$	$\{1,3\}$	$\{2,3\}$	$\{1\}$
$\{3\}$	$\varnothing,\{1,2\}$	$\{1,2,3\}$	$\varnothing$

Proof. A tiling determines its cut states uniquely. Conversely, place a horizontal domino at every row specified by each internal cut state, and fill the remaining cells in each column with the unique vertical completion. Condition (2.1) prevents overlapping horizontal dominoes and ensures that every remaining cell is covered. The empty endpoint states exclude dominoes extending beyond the rectangle. These constructions are inverse to one another. The empty path gives the empty tiling when $w = 0$. □

2.2. **First returns and block classification.**

Definition 2.2. An internal vertical grid line crossed by no domino is a *fault*. A nonempty tiling of a three-row rectangle is an *indecomposable block* if it has no internal fault.

Lemma 2.3. *The indecomposable blocks are exactly C and A_k, B_k for $k \geq 1$. Here C is the width-two block consisting of three horizontal dominoes. The block A_k has width $2k$ and consists of:*

(1) vertical dominoes in columns 1 and $2k$, each covering rows 1 and 2;

(2) horizontal dominoes in row 3 on column pairs $(2j-1, 2j)$, where $1 \leq j \leq k$;

(3) horizontal dominoes in each of rows $1, 2$ on column pairs $(2j, 2j+1)$, where $1 \leq j < k$.

The block B_k is the horizontal-axis reflection $h(A_k)$.

Proof. By Lemma 2.1, a block is a first-return path from the empty state. Its first step has three possibilities. If it reaches $\{1,2,3\}$, the next step is forced back to $\varnothing$, giving C. If it reaches $\{3\}$, every nonterminal excursion consists of the two steps $\{3\} \to \{1,2\} \to \{3\}$. After $k-1$ such excursions, the path returns from $\{3\}$ to $\varnothing$. This is a path of length $2k$ and gives exactly the stated coordinates of A_k. If the first step reaches $\{1\}$, the corresponding excursions are $\{1\} \to \{2,3\} \to \{1\}$, giving B_k. The complete transition table allows no other first-return path.

Every listed block is indeed indecomposable. In A_k, every odd internal cut is crossed by a domino in row 3, and every even internal cut is crossed by dominoes in rows $1, 2$. Reflection gives the same assertion for B_k. The unique internal cut of C is crossed in all three rows. The position of the first vertical domino distinguishes A_k from B_k, whereas C has no vertical domino. Thus the classification has neither omissions nor repetitions. □

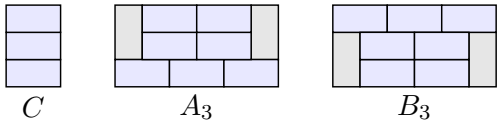


FIGURE 1. Representative blocks C, A_3, B_3. Rectangle boundaries are domino boundaries. The diagram was drawn from the coordinate definitions using AI-assisted TikZ code.

Lemma 2.4. *Every three-row tiling has a unique decomposition into a word over $\mathcal{B} = \{C\} \cup \{A_k, B_k : k \geq 1\}$. Let σ fix C and interchange A_k, B_k. Under this correspondence,*

$$h(W) = \sigma(W), \qquad v(W) = \mathrm{rev}(W), \qquad \rho(W) = \sigma(\mathrm{rev}(W)), \tag{2.2}$$

where σ is applied letterwise.

Proof. Cut a tiling at all its faults. The resulting pieces have no internal faults and hence are precisely the blocks of Lemma 2.3. Every boundary in a block decomposition must be a fault, and every fault must be used: omitting one would leave a decomposable piece. This proves uniqueness. Concatenation gives the inverse map, with the empty word corresponding to the empty tiling.

Horizontal-axis reflection acts on individual blocks by σ. Every block is invariant under vertical-axis reflection. For A_k, the column reversal $c \mapsto 2k+1-c$ interchanges the endpoint vertical dominoes. It sends each bottom-row pair $(2j-1, 2j)$ to another such pair, with index $k+1-j$, and each upper-row pair $(2j, 2j+1)$ to another, with index $k-j$. The block C is fixed domino by domino. Since the two reflections commute and $B_k = h(A_k)$, every B_k is also invariant under vertical-axis reflection.

Both reflections preserve the set of faults. Horizontal-axis reflection preserves block order, whereas vertical-axis reflection reverses it. Uniqueness of the decomposition now gives (2.2), including when block widths differ. □

2.3. **Fixed words.** Assign weight x to C and x^k to A_k, B_k. The exponent records half the width. The block generating function is

$$F(x) = x + 2\sum_{k\geq 1} x^k = \frac{3x - x^2}{1-x}. \tag{2.3}$$

Since $F(0) = 0$, only finitely many words contribute to any given coefficient.

Proposition 2.5. *The four fixed-point generating functions are*

$$D(x) = \frac{1}{1-F(x)} = \frac{1-x}{1-4x+x^2}, \qquad H(x) = \frac{1}{1-x}, \tag{2.4}$$

$$V(x) = (1+F(x))D(x^2) = \frac{1+3x+x^2-x^3}{1-4x^2+x^4}, \tag{2.5}$$

$$R(x) = (1+x)D(x^2) = \frac{1+x-x^2-x^3}{1-4x^2+x^4}. \tag{2.6}$$

Proof. An arbitrary word contributes to $D = \sum_{\ell\geq 0} F^\ell$. A word fixed by h must consist entirely of letters fixed by σ. The only such letter is C, giving H.

A word fixed by v is a palindrome. It has a unique expression $U\,\mathrm{rev}(U)$ or $Ub\,\mathrm{rev}(U)$, where b is any block. The two copies of U contribute $D(x^2)$, and the possible central block contributes $1 + F(x)$.

A word fixed by ρ has a unique expression $U\sigma(\mathrm{rev}(U))$, or $Ub\sigma(\mathrm{rev}(U))$ with $\sigma(b) = b$. In the latter case $b = C$, so the central contribution is $1+x$. In both constructions the two cases are disjoint and the decomposition is reversible. A central block may cross the geometric midline; that line is not required to be a fault. □

First proof of Theorem 1.1. Burnside's formula and Proposition 2.5 give

$$A(x) = \frac{1}{4}\left[\frac{1-x}{1-4x+x^2} + \frac{1}{1-x} + \frac{2(1+2x-x^3)}{1-4x^2+x^4}\right]. \tag{2.7}$$

Put $P_3 = 1-4x+x^2$ and $Q_3 = 1-4x^2+x^4$. On taking the common denominator $(1-x)P_3Q_3$, its numerator is

$$\frac{(1-x)^2Q_3 + P_3Q_3 + 2(1+2x-x^3)(1-x)P_3}{4} = 1 - 3x - 4x^2 + 10x^3 - 4x^5 + x^6.$$

This proves (1.1). The denominator expands to

$$1 - 5x + x^2 + 19x^3 - 19x^4 - x^5 + 5x^6 - x^7.$$

The numerator has degree six, so comparison of coefficients of x^n for $n \geq 7$ gives (1.2). The coefficients of degrees zero through six give the initial values in the theorem. □

Remark 2.6. In [2, equation (26)], the variable z marks the full width, so the appropriate comparison is $\overline{T}(z,3) = A(z^2)$. After substitution into (2.7), the horizontal-reflection contribution $H(z^2)/4$ can be written as

$$\frac{1}{8}\left(\frac{1}{1+z}+\frac{1}{1-z}\right) = \frac{1}{4(1-z^2)},$$

Rearranging the other contributions gives the cited expression. In the three-row formulas of [3, equations (4)–(6)], z instead marks the number of dominoes; the corresponding substitution is $z^3 = x$.

3. Half-rectangle bijections

We now derive the fixed-point counts without using the block classification or the palindrome argument. The local construction is stated for any positive height.

3.1. **Partial tilings and a transfer matrix.** For $m, j \geq 1$ and $S \subseteq [m] = \{1, \ldots, m\}$, let $\mathcal{P}_m(j, S)$ be the set of partial tilings of an $m \times j$ rectangle that cover every cell except the cells in rows S of the last column. Each uncovered cell is regarded as the left half of a horizontal domino to be completed in the next column. For $j = 0$, only the empty object with $S = \varnothing$ is allowed. Write $q_j(S) = |\mathcal{P}_m(j, S)|$.

Index rows and columns of T_m by all subsets of $[m]$, and set

$$T_m[p,q] = \begin{cases} 1, & p \cap q = \varnothing \text{ and every consecutive run} \\ & \quad \text{in } [m] \setminus (p \cup q) \text{ has even length,} \\ 0, & \text{otherwise.} \end{cases} \tag{3.1}$$

An empty complement satisfies the condition. The cells in p complete the incoming horizontal dominoes; those in q remain as outgoing halves. Each remaining consecutive run has a vertical tiling if and only if its length is even, and then that tiling is unique. This follows by pairing the endpoint with its only available neighbour and proceeding along the run. Consequently, legal state paths and partial tilings determine one another uniquely.

Let e_0 be the unit column vector at the empty state, and let $t_m(j)$ count complete $m \times j$ tilings. Then

$$q_j^{\mathsf{T}} = e_0^{\mathsf{T}} T_m^j, \qquad t_m(j) = e_0^{\mathsf{T}} T_m^j e_0. \tag{3.2}$$

For an auxiliary height-zero term, set $T_0 = (1)$ and $t_0(j) = 1$.

3.2. **Boundary conditions imposed by symmetry.** Define $J(S) = \{m + 1 - r : r \in S\}$. Let $\mathbf{1}$ be the all-ones column vector, and let $\chi_m(S) = 1$ if $J(S) = S$, and 0 otherwise. Denote the fixed-point counts of the two reflections and the half-turn on an $m \times 2n$ rectangle by $h_{m,n}, v_{m,n}, r_{m,n}$, respectively.

Lemma 3.1. *For every $m \geq 1$ and $n \geq 0$,*

$$v_{m,n} = e_0^{\mathsf{T}} T_m^n \mathbf{1}, \qquad r_{m,n} = e_0^{\mathsf{T}} T_m^n \chi_m. \tag{3.3}$$

Proof. Suppose first that $n \geq 1$. Given $P \in \mathcal{P}_m(n, S)$, reflect its internal dominoes into the right half of the rectangle, and insert a horizontal domino joining (r, n) to $(r, n+1)$ for each $r \in S$. The internal dominoes in the two halves are disjoint, and the inserted dominoes cover precisely the remaining cells. The resulting tiling is fixed by vertical-axis reflection. Conversely, restrict any such fixed tiling to its left half and cut its midline-crossing dominoes. This recovers P, S uniquely, proving the first bijection and formula.

For the half-turn, rotate the internal dominoes of P into the right half. The unmatched rows on the right are $J(S)$. A domino crossing the midline must join two cells in the same row, so completion is possible if and only if $J(S) = S$. When this condition holds, inserting one crossing domino in each row of S gives a half-turn-invariant tiling. Restriction to the left half is again the inverse map. Equivalently, necessity follows because the half-turn sends a

crossing domino in row r to one in row $m+1-r$. The empty-object convention gives both formulas for $n=0$. □

Lemma 3.2. *If $m=2k+1$ with $k\geq 0$, then*

$$h_{m,n}=t_k(2n) \qquad (n\geq 0). \tag{3.4}$$

Proof. In a tiling fixed by horizontal-axis reflection, a vertical domino incident with the middle row would have a distinct reflected image sharing its middle-row cell. Such a domino is impossible. The middle row must therefore have its unique horizontal tiling. The top k rows form a complete tiling, and their reflection determines the bottom k rows. Conversely, every tiling of the top subrectangle gives a fixed tiling in this way. Restriction is the inverse. The conventions cover $k=0$ and $n=0$. □

3.3. **A two-dimensional recurrence for height three.** From (3.1), the states reachable from the empty state at height three are

$$\varnothing,\quad \{1\},\quad \{3\},\quad \{1,2\},\quad \{2,3\},\quad \{1,2,3\}.$$

The empty state leads to $\{1\},\{3\},\{1,2,3\}$; $\{1\}$ leads to $\varnothing,\{2,3\}$; $\{3\}$ leads to $\varnothing,\{1,2\}$; The remaining transitions are

$$\{1,2\}\longrightarrow\{3\},\qquad \{2,3\}\longrightarrow\{1\},\qquad \{1,2,3\}\longrightarrow\varnothing.$$

We now make the induction explicit. Put

$$E=\{\varnothing,\{1,2\},\{2,3\}\},\qquad O=\{\{1\},\{3\},\{1,2,3\}\}.$$

Every transition out of E ends in O, and every transition out of O ends in E. There is no exit to either of the other two states. This is a statement about the complete list of allowed exits, not a claim that every pair of states in the two sets is connected.

Induction statement and base case. For each $j\geq 0$, the support of q_{2j} is contained in E, and

$$q_{2j}(\varnothing)=\alpha_j,\qquad q_{2j}(\{1,2\})=q_{2j}(\{2,3\})=\beta_j.$$

Thus β_j counts one specified state; the two nonempty states together contribute $2\beta_j$. We do not assert that all states of E have positive counts. At width zero only the empty state occurs, so $(\alpha_0,\beta_0)=(1,0)$, including equality of the two zero counts.

One added column. Assume the statement at width $2j$. Grouping by the previous state gives

$$\begin{aligned} q_{2j+1}(\{1\}) &= q_{2j}(\varnothing)+q_{2j}(\{2,3\})=\alpha_j+\beta_j,\\ q_{2j+1}(\{3\}) &= q_{2j}(\varnothing)+q_{2j}(\{1,2\})=\alpha_j+\beta_j,\\ q_{2j+1}(\{1,2,3\}) &= q_{2j}(\varnothing)=\alpha_j. \end{aligned}$$

All other counts are zero because the exits from E have been exhausted. Each legal transition has exactly one local filling. Conversely, removing the added column, and cutting any horizontal domino crossing its left boundary, recovers a unique previous object and state. The source classes are therefore disjoint and exhaustive; no extra multiplicity is needed in these sums.

A second added column. All three states in O can end at the empty state. Only $\{3\}$ can end at $\{1,2\}$, and only $\{1\}$ can end at $\{2,3\}$. Consequently,

$$\begin{aligned} q_{2j+2}(\varnothing) &= q_{2j+1}(\{1\})+q_{2j+1}(\{3\})+q_{2j+1}(\{1,2,3\})\\ &= (\alpha_j+\beta_j)+(\alpha_j+\beta_j)+\alpha_j=3\alpha_j+2\beta_j,\\ q_{2j+2}(\{1,2\}) &= q_{2j+1}(\{3\})=\alpha_j+\beta_j,\\ q_{2j+2}(\{2,3\}) &= q_{2j+1}(\{1\})=\alpha_j+\beta_j. \end{aligned}$$

There are no other nonzero counts, since all exits from O lie in E. This proves the support statement and equality of the two counts at the next even width, completing the induction

for every $j \geq 0$. In column-vector form the recurrence is

$$\begin{pmatrix}\alpha_{j+1}\\ \beta_{j+1}\end{pmatrix} = \begin{pmatrix}3 & 2\\ 1 & 1\end{pmatrix}\begin{pmatrix}\alpha_j\\ \beta_j\end{pmatrix}, \qquad (\alpha_0, \beta_0) = (1, 0). \tag{3.5}$$

Reading the four coefficients. The first row counts arrivals at $\varnothing$; the second counts arrivals at the single target $\{1, 2\}$.

Entry	Two-column continuation counted
3	From $\varnothing$ back to $\varnothing$, through any of $\{1\}$, $\{3\}$, or $\{1, 2, 3\}$.
2	One return to $\varnothing$ from each of $\{1, 2\}$ and $\{2, 3\}$, giving two source classes of size β_j.
1	From $\varnothing$ through $\{3\}$ to $\{1, 2\}$.
1	From $\{1, 2\}$ through $\{3\}$ back to $\{1, 2\}$; the other nonempty state contributes nothing to this target.

In particular, the coefficient 2 does not mean that each object in a fixed nonempty state has two returns to the empty state. For example, $(\alpha_1, \beta_1) = (3, 1)$ gives $\alpha_2 = (3+1)+(3+1)+3 = 11$ and $\beta_2 = 3 + 1 = 4$. This example illustrates the induction; it is not its basis.

Proposition 3.3. *The ordinary counts $d_j = t_3(2j)$ satisfy*

$$d_0 = 1, \qquad d_1 = 3, \qquad d_{j+2} = 4d_{j+1} - d_j \quad (j \geq 0). \tag{3.6}$$

For every $j \geq 0$,

$$\begin{aligned} h_n &= 1 \quad (n \geq 0), & r_{2j} &= r_{2j+1} = d_j,\\ v_{2j} &= d_{j+1} - 2d_j, & v_{2j+1} &= d_{j+1}. \end{aligned} \tag{3.7}$$

In particular,

$$\begin{aligned} a(2j) &= \frac{d_{2j} + 1 + d_{j+1} - d_j}{4},\\ a(2j+1) &= \frac{d_{2j+1} + 1 + d_{j+1} + d_j}{4}. \end{aligned} \tag{3.8}$$

Proof. Equation (3.2) gives $d_j = \alpha_j$. The matrix in (3.5) satisfies $M^2 - 4M + I = 0$, proving (3.6). At an even step, the only reachable J-invariant state is $\varnothing$; at an odd step it is $\{1, 2, 3\}$. Lemma 3.1 therefore gives the two formulas for r. Summing all terminal counts gives $\alpha_j + 2\beta_j = d_{j+1} - 2d_j$ at an even step and $3\alpha_j + 2\beta_j = d_{j+1}$ at an odd step. Lemma 3.2 gives $h_n = t_1(2n) = 1$. Substitution in Burnside's formula proves (3.8). □

Second proof of Theorem 1.1. The recurrence (3.6) gives $D(x) = (1 - x)/(1 - 4x + x^2)$. Summing (3.7) over the two parities gives

$$H(x) = \frac{1}{1-x}, \qquad R(x) = (1 + x)D(x^2),$$

and

$$V(x) = (1 + x)\frac{D(x^2) - 1}{x^2} - 2D(x^2) = \frac{1 + 3x + x^2 - x^3}{1 - 4x^2 + x^4}.$$

The quotient denotes the unique coefficient shift: $D(t) - 1 = tE(t)$, where $E(t) = \sum_{j\geq 0} d_{j+1}t^j$. It does not treat t as a unit in the formal power-series ring. Burnside's formula now gives (2.7); the algebraic simplification and coefficient comparison in the first proof finish the argument. The combinatorial derivation here uses only the partial-tiling model and the half-rectangle bijections. □

TABLE 2. Tilings and orbits classified by exact stabilizer.

Stabilizer	Tilings	Orbit size	Orbits
G	1	1	1
$\{e, h\}$	0	2	0
$\{e, v\}$	$v_n - 1$	2	$(v_n - 1)/2$
$\{e, \rho\}$	$r_n - 1$	2	$(r_n - 1)/2$
$\{e\}$	$d_n - v_n - r_n + 1$	4	$(d_n - v_n - r_n + 1)/4$

4. Stabilizers and asymptotics

Theorem 4.1. *For every $n \geq 0$, the exact stabilizers of three-row tilings are distributed as in Table 2.*

Proof. By Lemma 3.2, the only tiling fixed by h is the tiling in which each of the three rows is tiled horizontally. It is also fixed by v and ρ. A tiling fixed by any two distinct nonidentity elements is fixed by their product, and hence by all of G. Thus every pairwise intersection, as well as the triple intersection, of the three nonidentity fixed sets is this single tiling.

There are consequently $v_n - 1$ tilings with exact stabilizer $\{e, v\}$, $r_n - 1$ with exact stabilizer $\{e, \rho\}$, none with exact stabilizer $\{e, h\}$, and $d_n - v_n - r_n + 1$ with trivial stabilizer. Since G is abelian, the stabilizer is constant on an orbit. Dividing by the corresponding orbit size gives the table. These five subgroups exhaust all subgroups of G. □

For example, when $n = 8$ the fixed-point counts are $d_8 = 29681$, $v_8 = 265$, and $r_8 = 153$. Among the 7525 orbits, 7316 have trivial stabilizer, 132 have stabilizer $\{e, v\}$, 76 have stabilizer $\{e, \rho\}$, and one has stabilizer G.

Theorem 4.2. *Let $\lambda = 2 + \sqrt{3}$. If a tiling is chosen uniformly from $\mathcal{T}_n$, let p_n be the probability that it has a nonidentity symmetry. Then, as $j \to \infty$,*

$$p_{2j} \sim (\lambda - 1)\lambda^{-j}, \qquad p_{2j+1} \sim \frac{\lambda + 1}{\lambda}\lambda^{-j}. \tag{4.1}$$

In particular, $p_n = O(\lambda^{-n/2}) \to 0$, and

$$a(n) = \frac{d_n}{4} + O(\lambda^{n/2}) \sim \frac{3 + \sqrt{3}}{24}\lambda^n. \tag{4.2}$$

Proof. Solving (3.6) with its initial values gives

$$d_n = c_+\lambda^n + c_-\lambda^{-n}, \qquad c_\pm = \frac{3 \pm \sqrt{3}}{6}. \tag{4.3}$$

By Table 2, $p_n = (v_n + r_n - 1)/d_n$, whereas (3.7) gives

$$v_{2j} + r_{2j} = d_{j+1} - d_j, \qquad v_{2j+1} + r_{2j+1} = d_{j+1} + d_j.$$

The leading terms of these numerators are $c_+(\lambda - 1)\lambda^j$ and $c_+(\lambda + 1)\lambda^j$, respectively. The corresponding denominators have leading terms $c_+\lambda^{2j}$ and $c_+\lambda^{2j+1}$. This proves (4.1). Finally, $a(n) = (d_n + 1 + v_n + r_n)/4$ gives (4.2). □

Uniform sampling of orbits gives a different probability space. The number of orbits with nontrivial stabilizer is $(v_n + r_n)/2$, so their proportion is $(v_n + r_n)/(2a(n))$, which also tends to zero.

5. Rectangles of arbitrary odd height

Fix $m = 2k+1 \geq 1$. Let $a_m(n)$ be the number of domino tilings of an $m \times 2n$ rectangle up to its full symmetry group, with $a_m(0) = 1$. A nonempty rectangle of odd height and even width is never a square, so its symmetry group is still $\{e, h, v, \rho\}$. Use T_m, χ_m, e_0 as defined in Section 3. The following is an explicit specialization of the general recurrence framework discussed by DeFord [1]. We include the proof to specify all symmetry boundary terms.

Theorem 5.1. *For every $m = 2k+1 \geq 1$ and $n \geq 0$,*

$$a_m(n) = \frac{1}{4}\left[e_0^{\mathsf{T}} T_m^{2n} e_0 + e_0^{\mathsf{T}} T_k^{2n} e_0 + e_0^{\mathsf{T}} T_m^{n}(\mathbf{1} + \chi_m)\right]. \tag{5.1}$$

Equivalently,

$$\begin{aligned} A_m(x) := \sum_{n\geq 0} a_m(n) x^n = \frac{1}{4}\big[&e_0^{\mathsf{T}}(I - xT_m^2)^{-1} e_0 \\ &+ e_0^{\mathsf{T}}(I - xT_k^2)^{-1} e_0 \\ &+ e_0^{\mathsf{T}}(I - xT_m)^{-1}(\mathbf{1} + \chi_m)\big]. \end{aligned} \tag{5.2}$$

For each fixed odd height, $A_m(x)$ is rational, and its coefficient sequence satisfies a constant-coefficient linear recurrence.

Proof. The identity contribution is given by (3.2). Lemma 3.2 gives the horizontal-axis reflection contribution, and Lemma 3.1 gives the other two. The same double count as in Lemma 1.2 proves (5.1). Summing over n and using $\sum_{n\geq 0} x^n M^n = (I - xM)^{-1}$ proves (5.2). Every matrix is finite, and $I - xM$ has constant term I. Its inverse exists formally, and its entries are rational by the adjugate formula. Multiplying the resulting generating function by a denominator with nonzero constant term and comparing sufficiently high coefficients gives the asserted recurrence. □

The vectors e_0 in the different summands belong to the corresponding state spaces; when $k = 0$, the convention $T_0 = (1)$ applies. Lemma 3.1 itself holds for even heights as well. However, (3.4) uses a middle row and cannot be applied unchanged at even height. That case requires a separate treatment of vertical dominoes crossing the horizontal axis. For a square, quarter-turns and diagonal reflections must also be included to count classes under the full symmetry group. Neither extension is asserted here.

6. An explicit formula for height five

At height five the transfer calculation reduces to a six-dimensional polynomial certificate. We give the reduced matrix and boundary vectors explicitly; Appendix A supplies the single-column matrices from which they are obtained.

Represent a row subset by a five-bit mask, with the least significant bit corresponding to the top row. The row-reversal orbits of reachable even-column states, in the chosen order, are

$$\{0\},\ \{3, 24\},\ \{6, 12\},\ \{9, 18\},\ \{15, 30\},\ \{27\}. \tag{6.1}$$

Row-reversed states have equal counts. Let each component of u_j record the count of *one individual state* in the corresponding orbit after $2j$ columns, rather than the sum over that orbit. Then $u_0 = e_0$ and $u_{j+1} = Mu_j$, where

$$M = \begin{pmatrix} 8 & 6 & 4 & 2 & 2 & 1 \\ 3 & 4 & 1 & 1 & 1 & 1 \\ 2 & 1 & 2 & 1 & 1 & 0 \\ 1 & 1 & 1 & 2 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}. \tag{6.2}$$

Define

$$P(t) = 1 - 15t + 32t^2 - 15t^3 + t^4, \qquad N(t) = \begin{pmatrix} 1 - 7t + 7t^2 - t^3 \\ 3t - 4t^2 + t^3 \\ 2t - 5t^2 + t^3 \\ t + t^2 \\ t - t^3 \\ t - t^3 \end{pmatrix}. \tag{6.3}$$

Polynomial multiplication gives the exact identity

$$(I - tM)N(t) = P(t)e_0. \tag{6.4}$$

Since $P(0) = 1$, it follows that

$$f(t) := \sum_{j \geq 0} u_j t^j = \frac{N(t)}{P(t)}.$$

Appendix A also records all four nonconstant coefficient-vector identities underlying (6.4).

The terminal row vectors for vertical-axis reflection are

$$w_{v,e} = (1, 2, 2, 2, 2, 1), \qquad w_{v,o} = (8, 6, 4, 4, 2, 1),$$

and those for the half-turn are

$$w_{r,e} = (1, 0, 0, 0, 0, 1), \qquad w_{r,o} = (2, 2, 0, 0, 0, 1).$$

The even vectors sum all states or only row-reversal-fixed states. The odd vectors first apply one column transition and then take the appropriate sum. Lemma 3.1 gives

$$\begin{aligned} V_5(x) &= (w_{v,e} + x w_{v,o}) f(x^2), \\ R_5(x) &= (w_{r,e} + x w_{r,o}) f(x^2). \end{aligned} \tag{6.5}$$

Theorem 6.1. *The four fixed-point generating functions at height five are*

$$D_5(x) = \frac{1 - 7x + 7x^2 - x^3}{P(x)}, \qquad H_5(x) = \frac{1 - x}{1 - 3x + x^2}, \tag{6.6}$$

$$V_5(x) = \frac{1 + 8x + 8x^2 - 23x^3 - 9x^4 + 16x^5 - x^7}{P(x^2)}, \tag{6.7}$$

$$R_5(x) = \frac{1 + 2x - 6x^2 - 7x^3 + 7x^4 + 6x^5 - 2x^6 - x^7}{P(x^2)}. \tag{6.8}$$

The orbit generating function is

$$A_5(x) = \frac{D_5(x) + H_5(x) + V_5(x) + R_5(x)}{4}. \tag{6.9}$$

Proof. A complete tiling has empty terminal state, so $D_5 = f_0$, giving the first formula. Substitution of (6.3) into (6.5) gives the two formulas for V_5, R_5.

By Lemma 3.2, $h_{5,n} = t_2(2n)$. A two-row tiling either starts with one vertical domino or with two horizontal dominoes covering its first two columns. Thus $t_2(j) = t_2(j-1) + t_2(j-2)$ for $j \geq 2$, with initial values $t_2(0) = t_2(1) = 1$. The two-step recurrence matrix is $L = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, which satisfies $L^2 - 3L + I = 0$. Hence $c_n = t_2(2n)$ satisfies $c_{n+2} = 3c_{n+1} - c_n$, with $c_0 = 1, c_1 = 2$, proving the formula for H_5. Finally, Burnside's formula proves (6.9). □

Write

$$Q(x) = (1 - 3x + x^2)P(x)P(x^2) = \sum_{j=0}^{14} q_j x^j, \qquad U(x) = \sum_{j=0}^{13} b_j x^j,$$

where $A_5(x) = U(x)/Q(x)$. Table 3 lists all coefficients and the fourteen initial values needed for

$$a_5(n) = -\sum_{j=1}^{14} q_j a_5(n-j), \qquad n \geq 14. \tag{6.10}$$

No minimality assertion about the recurrence order is needed.

TABLE 3. Coefficients of U, Q and initial values; $b_{14} = 0$.

j	b_j	q_j	$a_5(j)$
0	1	1	1
1	-13	-18	5
2	6	63	33
3	194	144	329
4	-365	-1060	3818
5	-371	1296	46878
6	1304	1312	584386
7	-594	-3492	7318152
8	-703	1312	91752831
9	717	1296	1150774799
10	-154	-1060	14434490195
11	-36	144	181060839597
12	15	63	2271175917429
13	-1	-18	28489049801109
14	0	1	by recurrence

The values $a_5(1), \ldots, a_5(5) = 5, 33, 329, 3818, 46878$ agree with the entries for widths $2, 4, 6, 8, 10$ in [2, Table 12]. This is a comparison made after the derivation; those values were not used to fit a matrix or a generating function.

7. Formal evidence for the three-row theorem

The M1–M6 development follows Section 3, without using the block classification. A cell has zero-based coordinates in `Fin m` × `Fin w`. A domino is an unordered two-element set of adjacent cells, and a tiling is an exact cover. The definition `Domino.a n` is the cardinality of the quotient of actual $3 \times 2n$ tilings by the four coordinate symmetries. The coefficient of X^n in `orbitSeries` is $(a(n) : \mathbb{Q})$; neither the target recurrence nor the rational expression defines it.

Stage	Mathematical interface to examine
M1	Unordered dominoes, exact coverage, terminal holes, the width-zero guard, coordinate action, fixed sets, and the orbit quotient.
M2	An equivalence between actual partial tilings and state paths, with two inverse identities, followed by the counting recurrence.
M3	Reflection and half-turn completion/restriction, including actual internal and crossing dominoes; uniqueness of the h-fixed tiling.
M4	Simultaneous induction of support and counts for all widths; identification of the auxiliary pair with actual cardinalities.
M5	Burnside applied to that action, divisibility, and the empty and small-board cases; no faithfulness assumption.
M6	Series defined from those counts, denominator identity and invertibility, the recurrence, and all seven initial values.

Write $B = (1-X)(1-4X+X^2)(1-4X^2+X^4)$ and $N = 1-3X-4X^2+10X^3-4X^5+X^6$. The development first proves $BA = N$, then uses $B(0) = 1$ to obtain $A = NB^{-1}$ in $\mathbb{Q}[[X]]$.

Coefficient comparison at $n+7$ gives the recurrence for every $n \in \mathbb{N}$, interpreted in $\mathbb{Z}$; a separate natural-number version moves all negative terms to the other side. Coefficients zero through six give the initial values, and the first recurrence step gives $a(7) = 2037$. The main endpoints, in namespace `Domino`, are `orbit_series_paper`, `orbit_recurrence_seven`, `orbit_initial_values`, and `three_row_main_theorem`.

The recorded October 25, 2026 acceptance uses Lean 4.35.0 and the following mathlib commit:

`56a005855ea17e0b0958f73ec801ae140b0a1679`.

It covers 32 mathematical modules and audits 413 explicit declarations, including auxiliary definitions and instances. The allowed axioms are `propext`, `Classical.choice`, and `Quot.sound`. Project sources were rebuilt in a directory without prior project build products, reusing the checked third-party dependency cache; this was not a cache-free rebuild of all of mathlib.

This edition deliberately fixes its review scope at M1–M6. The general-height statements, stabilizers, asymptotics, five-row certificate, and block-based second proof are not certified by that six-stage evidence. Later local stages are separate work. The four coordinate transformations are explicit; a classification of all Euclidean isometries is not formalized. At zero width the abstract group acts trivially on the unique empty tiling. Kernel checking does not decide whether these definitions express the intended question, or establish bibliographic priority.

The accompanying `review/m1-m6/` directory provides a source-pinned semantic map, a dependency guide, executable declaration probes, and an unfilled external review form. It distinguishes the compiler checks from the human judgments still requested. In particular, the pair in (3.5) is defined recursively in Lean and only then proved equal to the geometric counts by `profile_even_odd`; the existence of its recursion equations alone would not establish the tiling count.

8. Computational checks and further questions

The formal evidence for the three-row main theorem is described in Section 7. Other arguments lie outside the M1–M6 scope reported in this edition. Computation checks implementations, transcriptions, and specified algebraic identities. It does not complete the research. The block-based verification enumerated all three-row tilings for $0 \le n \le 8$, a total of 40545 tilings, and checked factorization, reconstruction, and all four symmetry actions. It reused an earlier row-major grid enumerator. A separate column-major enumerator, selecting at each step an unmatched cell with the fewest available neighbours, was subsequently implemented without importing the old enumerator or the block code. Its exhaustive ranges are shown in Table 4.

Table 4. Ranges of the second exhaustive enumeration.

Height	Half-width n	Main checks
1	$0, \dots, 8$	Smallest height and nonfaithful actions
3	$0, \dots, 8$	Orbit counts and exact stabilizers
5	$0, \dots, 4$	Five-row formulas and midline completion
7	$0, \dots, 2$	Higher odd-height instances

The second enumerator examined 57468 tilings in total. For every tiling it computed the four cellwise symmetry images and a canonical orbit representative. For vertical-reflection and half-turn fixed tilings, it restricted to the left half, recorded the terminal row set, and reconstructed the full tiling. The resulting counts for each terminal state agreed with the transfer computation, and the three-row stabilizer counts agreed with Table 2.

Exact symbolic calculations also solved the full three-row and five-row matrix systems. Each resolvent row was checked against its defining linear equation with zero residual. The six-dimensional five-row model was checked against the full matrix model and against (6.4). For both heights, coefficients of the rational functions agreed with integer transfer iteration for $0 \le n \le 100$. This last comparison was not exhaustive tiling enumeration over that range. The recorded research runs used Python 3.13.0 and SymPy 1.13.1 on Windows on October 25, 2026. They are deterministic and use exact arithmetic. Commands, input ranges, source hashes, and outputs are included in the supplementary research materials.

The two proof routes share the local column-covering model but use different global bijections. They were developed in the same AI-assisted workflow, as disclosed below. Agreement of the implementations does not replace independent mathematical review. In particular, finite enumeration does not prove the unbounded-parameter statements; these follow from the bijections. The polynomial certificate establishes the evaluation of a specified matrix model.

For even height, the horizontal-axis reflection term must allow dominoes crossing that axis. For squares, the full dihedral group requires two quarter-turns and two diagonal reflections in addition to the four symmetries considered here. These provide natural extensions, but no formula for those cases is asserted in this paper.

Appendix A. The reduced five-row certificate

The even-column state orbits are listed in (6.1). The odd-column orbits, in order, are

$$\{1,16\},\ \{4\},\ \{7,28\},\ \{13,22\},\ \{19,25\},\ \{31\}.$$

These twenty states form the set reachable from the empty state. This can be checked directly from (3.1): starting at mask zero and listing all successors reaches exactly these states, and all their successors remain in the set. Row reversal commutes with the transition relation and fixes the initial state, so reversed states have equal counts.

Let u_j be the even-column count vector, with one-state counts as in Section 6, and let z_j be the next odd-column vector. Then $z_j = Bu_j$ and $u_{j+1} = Cz_j$, where

$$B = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}. \tag{A.1}$$

For an input orbit O_j and an output representative q_i, the corresponding entry is

$$\sum_{p \in O_j} T_5[p, q_i].$$

It does not depend on the choice of representative. For example, $B_{2,2} = 2$ because both masks 3 and 24 can precede mask 4. The reduced matrices need not be symmetric: their variables record the count of each state, not the total of an orbit. Matrix multiplication gives $CB = M$.

The even orbit-size vector is $(1,2,2,2,2,1)$ and the odd orbit-size vector is $(2,1,2,2,2,1)$. The fixed masks of row reversal are 0 and 27 on the even side, and 4 and 31 on the odd side. Consequently,

$$w_{v,o} = (2,1,2,2,2,1)B, \qquad w_{r,o} = (0,1,0,0,0,1)B,$$

which gives the terminal vectors in the text.

To check (6.4) coefficientwise, write $N(t) = n_0 + n_1 t + n_2 t^2 + n_3 t^3$, with

$$\begin{aligned} n_0 &= (1,0,0,0,0,0)^{\mathsf{T}}, & n_1 &= (-7,3,2,1,1,1)^{\mathsf{T}}, \\ n_2 &= (7,-4,-5,1,0,0)^{\mathsf{T}}, & n_3 &= (-1,1,1,0,-1,-1)^{\mathsf{T}}. \end{aligned}$$

Direct integer multiplication gives

$$n_1 - Mn_0 = -15e_0, \quad n_2 - Mn_1 = 32e_0,$$
$$n_3 - Mn_2 = -15e_0, \qquad -Mn_3 = e_0.$$

Together with $n_0 = e_0$, these are all coefficients of $(I - tM)N(t) = P(t)e_0$. Thus the certificate consists of finitely many explicit integer identities and determines every coefficient of the generating functions.

Appendix B. Reproducibility and supplementary materials

The accompanying public repository includes selected problem, approach, lemma, proof, and source notes in Chinese and English, together with verification instructions. Its `research/02_problems/P002-domino-orbit-counting/` directory preserves the original dependency layout. The historical grid enumerator is in `research/01_topics/checks/a232622-screening/`. Historical numerical reports are published with local paths, invocation details, and platform metadata removed. Their numerical fields are unchanged, and each report identifies this treatment. These sanitized records are distinct from a new experiment. The six selected research programs are unchanged. Private archives, personal metadata, and machine logs are not included.

Relative to the problem directory, the principal programs are:

- `attempts/A001-irreducible-blocks/checks/verify_blocks.py` for block factorization and the four symmetry actions;
- `attempts/A002-half-board-transfer/checks/transfer_core.py` for the column model and integer fixed-point counts;
- `attempts/A002-half-board-transfer/checks/derive_functions.py` for exact rational functions and resolvent certificates;
- `attempts/A002-half-board-transfer/checks/verify_geometry.py` for the second exhaustive enumeration and half-rectangle reconstruction;
- `attempts/A002-half-board-transfer/checks/five_row_certificate.py` for the reduced five-row certificate.

The programs use Python and SymPy. The first verification program also imports the historical enumerator by its relative location, so the package directory structure must be retained. Exact commands and environment information appear in the English verification records. Executing a historical program can regenerate its report; the supplied reports should be retained when reproducing the computations in a working copy.

The English manuscript has a separate audit that compares its displayed rational functions, matrices, terminal vectors, and polynomial certificate with the research records. The five-row coefficient table is generated from the exact recorded numerator and denominator, and its coefficients are compared with integer matrix iteration. This audit is documented separately from the historical exhaustive enumerations. The LaTeX source, build script, and file-hash manifest are included for independent examination.

Use of AI tools

OpenAI Codex was used on October 24–25, 2026, to assist with the mathematical derivations, source searches, verification programs, diagram code, drafting, and English translation. The two proof routes and their computational checks were developed within the same AI-assisted workflow; their agreement does not constitute independent review. No claim of completed independent refereeing is made. Machine verification is reported only for the M1–M6 scope in Section 7. The accompanying research records specify the checks actually performed and their scope.

References

[1] D. DeFord, *Enumerating distinct chessboard tilings*, Fibonacci Quart. **52** (2014), no. 5, 102–116. https://www.fq.math.ca/Papers1/52-5/DeFord.pdf.

[2] R. J. Mathar, *Paving rectangular regions with rectangular tiles: tatami and non-tatami tilings*, preprint (2013), arXiv:1311.6135. https://arxiv.org/abs/1311.6135.

[3] R. J. Mathar, *Tilings of rectangular regions by rectangular tiles: Counts derived from transfer matrices*, preprint (2014), arXiv:1406.7788. https://arxiv.org/abs/1406.7788.

[4] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A232622, Number of incongruent domino tilings of the $3 \times (2n)$ board, https://oeis.org/A232622, accessed October 25, 2026.

[5] R. P. Stanley, *Topics in Algebraic Combinatorics*, lecture notes, version of February 1, 2013. https://math.mit.edu/~rstan/algcomb/algcomb.pdf.